Geophys. J. Int. (1991) **104**, 635–646

Palaeomagnetism of Devonian ring complexes from the Bayuda Desert, Sudan—new constraints on the apparent polar wander path for Gondwanaland

V. Bachtadse[1]* and J. C. Briden[1,2]

[1] *Department of Earth Sciences, Oxford University, Parks Road, Oxford* OX1 3PR, *UK*
[2] *Natural Environment Research Council, Polaris House, North Star Avenue, Swindon* SN2 1EU, UK

Accepted 1990 November 4. Received 1990 October 23; in original form 1990 April 27

SUMMARY

The approach of Gondwana, and/or microplates derived from its northern margins, towards Laurentia and the closure of the intervening Palaeo-Tethys is a controlling factor in the geodynamic evolution of the Atlantic bordering continents during the early and mid-Palaeozoic. However the currently available palaeomagnetic data for Gondwana, especially for the geodynamically critical Silurian to Devonian interval are still rather sparse and are compatible with a number of palaeogeographic scenarios. In order to address this issue 114 samples (18 sites) have been collected from a series of high level acidic intrusive rocks, rhyolitic lavas and tuffs of mid-Devonian age (377 ± 5 Ma, Rb/Sr) along the southern escarpment of the Gilif Hills, Sudan (17.83°N, 32.67°E). Detailed demagnetization experiments revealed the presence of three directionally distinct components of magnetization, labelled **A**, **B** and **C**. The overall mean direction of component **A** (*Dec.* 0.9°, *Inc.* 29.8°, $\alpha_{95} = 4.6°$, $k = 91.1$, $N = 12$ sites) is indistinguishable from either the direction of the present field or the geocentric axial dipole field in northern Sudan. Mixed polarities suggest a Recent or Pleistocene age of this component. Component **B** also displays mixed polarities. The site mean directions for this component average to 6.5° declination, −40.2° inclination with $\alpha_{95} = 7.0°$ and $k = 48.1$ (based on 10 sites) and yield a palaeomagnetic pole at 48.8°S, 23.5°E ($dp = 5.1°$, $dm = 8.4°$). Maghaemite and haematite have been identified to be the predominant magnetic phases in samples which carry component **A** and/or **B**. Both **A** and **B** are the probable results of low-temperature oxidation of magnetite. Although the age of component **B** can not be directly determined, it is inferred by correlation with published data to reflect a late Carboniferous overprint. Component **C**, identified in 11 sites (39 samples), is characterized by a very steep downward magnetization (*Dec.* 296.2°, *Inc.* 79.3°, $\alpha_{95} = 10.8°$, $k = 18.7$, $n = 11$ sites) which corresponds to a palaeomagnetic pole position in the southwestern corner of Libya (25.9°N, 11.6°E, $dp = 19.6°$, $dm = 20.6°$). Maximum blocking temperatures of component **C**, generally below 580 °C, are indicative of magnetite as carrier of this remanence. The results of thermal remanent magnetization experiments demonstrate that component **C** is a thermo-remanent magnetization, probably primary, and suggest this component to be representative for the direction of the Earth's magnetic field during emplacement of the Gilif Hills volcanic rocks in mid-Devonian times. The position of the resulting palaeomagnetic pole lends new support to geodynamic scenarios advocating the existence of a wide (approx. 5000 km) Pre-Hercynian ocean during Devonian times. The simplest reconciliation of component **B** with the reference apparent polar wander path is achieved by assuming it to be of Carboniferous age

* New address: Inst. f. Allg. & Angew. Geophysik, Ludwig-Maximillians-Universität, Theresienstr. 41, D-8000 München 2, Germany.

and by invoking a westward cusp in the Carboniferous part of the apparent polar wander path. Taken at face value, the age of this cusp coincides with large-scale post-orogenic right lateral strike slip movements in Hercynian Europe and North Africa. This interpretation does not require the apparent polar wander (APW) loop and very high rates of plate motion implied by some alternative models. However, if it is valid it will require reinterpretation of the remanence-age of one West African intrusive suite and of some eastern Australian Palaeozoic rocks.

Key words: Africa, apparent polar wander path, Carboniferous, Devonian, Gondwana, ring complexes.

1 INTRODUCTION

The convergence of Gondwana to Laurasia is a key factor dictating the tectonic pattern for Euramerica and northern Gondwana during the whole of the Palaeozoic. Palaeozoic continental reconstructions, however, are still too poorly constrained for use in geodynamic interpretations of the development of the Palaeozoic Atlantic bordering orogenies. Based upon palaeoclimatological evidence (see reviews by Caputo & Crowell 1985; Veevers & Powell 1987; Scotese & Barrett 1990) and on the few reliable palaeomagnetic data for Gondwana (e.g. reviews by Brock 1981; Livermore, Smith & Briden 1985; Van der Voo 1988; Bachtadse & Briden 1990) the palaeo-south pole for Gondwana shifted from a position west of Northern Africa in the Ordovician to a Carboniferous location east of South Africa (Fig. 1). However, there remains a considerable controversy regarding the exact timing and the nature of this shift.

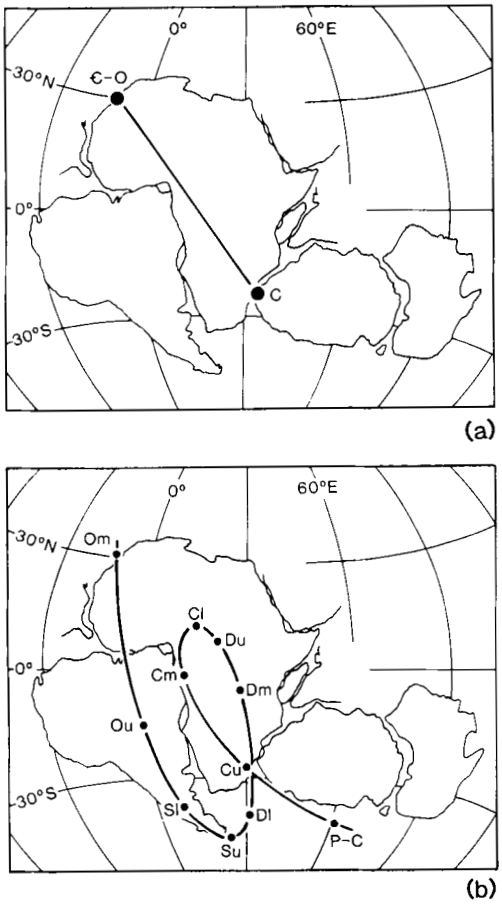


Figure 1. Alternative apparent polar wander paths for Gondwana during the Palaeozoic. (a) McElhinny *et al.* (1968); (b) Bachtadse & Briden (1990).

The palaeomagnetic argument centres around the question whether the well-established mean palaeo-south poles for the Ordovician and Permian should be linked up in a rather straight line (Fig. 1a; McElhinny *et al.* 1968) or whether there is sufficient evidence for a more complex APW path such as that in Fig. 1(b) (Bachtadse & Briden 1990). Paths of the latter kind were originally proposed by Morel & Irving (1978) exclusively on the basis of Australian data and more recently by Hargraves, Dawson & Van Houten (1987) including data from the Aïr in Niger, and by Li *et al.* (1990). Translated into plate tectonic terms, this controversy is between straightforward closure of an ocean between two formerly widely separated supercontinents (Gondwana and Laurasia), and the interruption of this convergence by a temporary divergence of 3000 km or more. The latter hypothesis predicts the repeated closure and opening of the proto-Tethys but requires drift rates in excess of 25 cm yr^{-1} over a period of *c.* 70 Myr—faster then any drift rate for any tectonic plate in post-Palaeozoic time.

The solution to this critical question remains elusive but must depend on palaeomagnetic data constraining the drift history of Gondwana during the early Silurian to the late Devonian.

The key factors are the reliability and the *relative* ages of the remanence in (a) various Devonian formations that yield poles in Central Africa and (b) ring complexes in the Aïr, to which ages of 435 Myr have been assigned (Hargraves *et al.* 1987), which give a pole in the South Atlantic. Only if these Devonian poles are valid *and* the Aïr pole really pre-dates them, will the more complex APW path and the faster drift rates be established.

(a) For more than a decade the palaeomagnetic result from the Msissi Norite of Morocco (Hailwood 1974), yielding a palaeo-south pole in Central Africa, has been considered as the only reliable Late Devonian palaeomagnetic pole for Africa. However, recent isotopic age determinations (K/Ar) on the Msissi Norite, have yielded a Jurassic age and the original palaeomagnetic results have therefore been judged unreliable (Salmon *et al.* 1986).

Subsequent palaeomagnetic studies on folded carbonates of Devonian age from western Algeria and Morocco

(Salmon *et al.* 1988; Aïfa, Feinberg & Pozzi 1990), flat lying Palaeozoic clastic sediments from Mauritania (Kent, Dia & Sougy 1984) and volcanic rocks from the Sabaloka ring complex, Sudan (Soffel *et al.* 1990) yield a fairly tight cluster of palaeomagnetic south poles, conspicuously close to the location of the Permo/Carboniferous palaeo-south pole, located to the southeast of Southern Africa. The directions reported in the Moroccan and Algerian studies either fail fold tests (Salmon *et al.* 1988) or point towards a synfolding age of remagnetization (e.g. Carboniferous) (Aïfa *et al.* 1990). No field tests of palaeomagnetic stability have been applied to the results from Mauritania and Sudan and the proximity of these two poles to the Permo/Carboniferous reference pole favours a secondary magnetization. This leaves only palaeomagnetic results from sediments from the Devonian Bokkeveld Group, South Africa, (BO in Table 2 and Fig. 10; Bachtadse, Van der Voo & Haelbich 1987), as indicative that the African south pole was located in Central Africa during the Devonian and those data are of rather poor quality.

(b) The palaeomagnetic pole for the Aïr ring complexes (Hargraves *et al.* 1987) is altogether more soundly based, on thermal and AF demagnetization of 247 samples from 38 sites covering 12 individual ring complexes of acidic and basic composition. Results from 15 sites, 68 samples from six ring structures yielded northerly and upward pointing magnetizations (Group A). Rather steep (>65′) directions, pointing to the west have been reported from four sites (16 samples) from three ring complexes, whereas the remainder of the collection yielded magnetizations either very close to the present direction of the geocentric axial dipole field or too scattered to calculate site mean directions. Hargraves *et al.* (1987) argued that 'there is no petrographic evidence to suggest that the Group A vector is not a primary TRM or TCRM'. Yet there is still room for unease at this result: whether the age of 435 Myr derived by Ar^{40}/Ar^{39} dating of biotite from two of the palaeomagnetically sampled complexes fully reflects the age of the Group A magnetization, given (i) that Rb/Sr ages from the Aïr complexes follow a linear trend from 480 Myr in the north to 374 Myr in the south (Bowden *et al.* 1976), and that the site mean directions have a great circle distribution if the results from Tamgak and Abontorok are included (Hargraves *et al.* 1987, Fig. 6b). Furthermore, directions close to the PDF were interpreted as either the result of weathering or an effect of extensive Quaternary volcanism in the southern Aïr, and no attempt was made to interpret the steep westerly NRM component which, as we will see later, recurs in the Sudan rocks that are the subject of our present study.

Hence the case for the loop in the African APW path and the opening/closing phase that it implies, depends (a) on the location of the Gondwana Devonian pole in Central Africa, which currently hangs almost entirely on data from Australia and (b) the age of the remanence in the Aïr rocks. In this paper we report the results of a palaeomagnetic study on mid to late-Devonian volcanic rocks from the Bayuda Desert in Sudan, which address the first of these issues.

2 GEOLOGICAL SETTING AND SAMPLING

Anorogenic volcanic rocks were emplaced throughout the Phanerozoic over the entire African and Arabian Plates (see review by Kinnaird & Bowden 1987). The erosionally exposed roots of those volcanoes and their cauldrons are represented by a wide variety of igneous ring complexes ranging from annular to elliptical with diameters ranging from less than 1 km to 30 km or more. Their petrological composition ranges from carbonatite, gabbro and syenite to

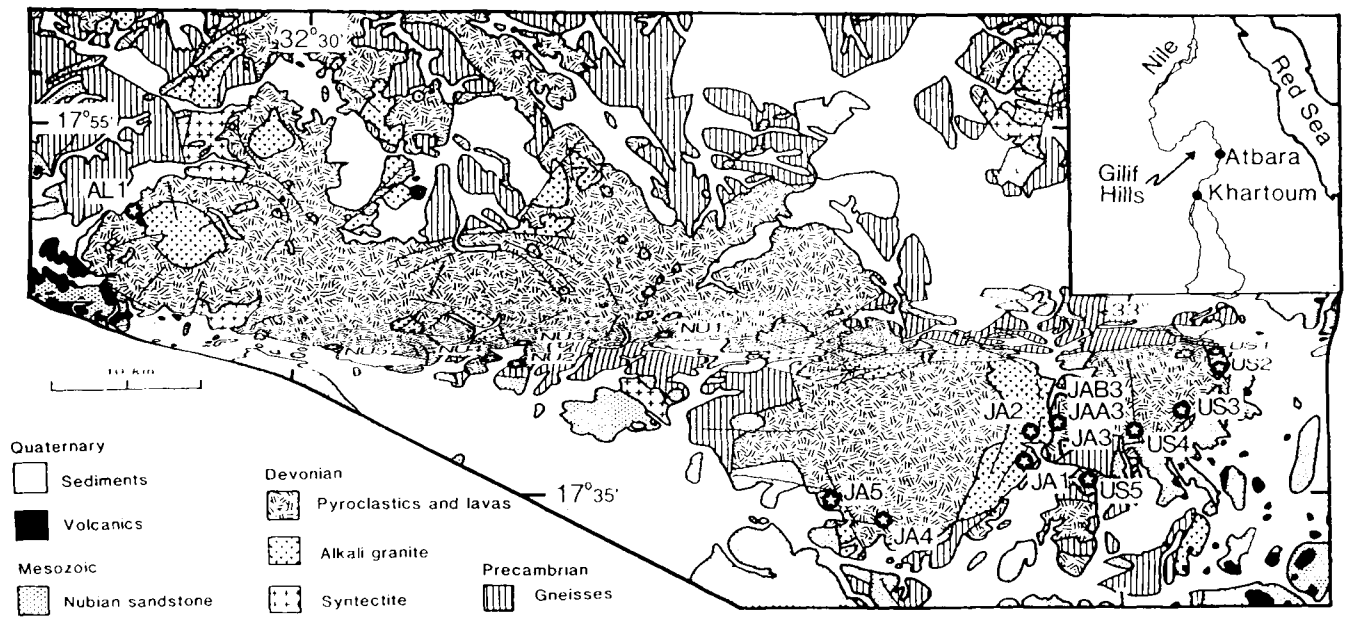


Figure 2. Geological sketch map of the Gilif Hills, Sampling sites are marked as stars.

granite and their effusive equivalents (see Kinnaird & Bowden 1987).

So far the occurrence of more than 100 ring complexes has been reported from northeast Africa, where they intrude the metamorphic Precambrian basement and also Palaeozoic and Mesozoic sediments (Vail 1976, 1989).

The igneous ring complexes of the Bayuda Desert in Northern Sudan have been the subject of a comprehensive petrological and geochemical investigation (Barth *et al.* 1983). Detailed (Rb/Sr) whole rock isochron dating on 13 individual igneous ring complexes yielded ages from 574 to 157 Myr indicating at least five pronounced episodes of volcanic activity in the region (Barth *et al.* 1983). The Gilif Hills complexes (Fig. 2) in the southern part of the Bayuda Desert show a rather high erosional level and display composite structure of intersecting ring complexes, dominated by alkali granite and alkali rhyolitic extrusives (Barth *et al.* 1983). Members of the intrusive as well as the effusive phase of the Gilif Hills complexes have been dated isotopically and the time of emplacement has been well established to be 377 ± 5 Myr (Rb/Sr decay constant 1.42×10^{-11} yr^{-1}; Barth *et al.* 1983).

The Gilif Hills are surrounded by flat-lying Mesozoic sandstones suggesting no Mesozoic and/or post Mesozoic tilting of the region. In addition the shape and outcrop pattern of the intrusives and the flows themselves also indicate that the complexes are still in their original position.

A total of 114 drill cores were collected using a portable petrol powered drill at 18 sites from five individual ring complexes along the southern margin of the Gilif Hills (Fig. 2). A standard sun compass was used to orient the samples.

3 PALAEOMAGNETIC METHODS

Standard palaeomagnetic cylinders (25 mm diameter and 23 mm long) were prepared from the oriented core samples. Direction and intensity of their remanent magnetization was measured on a two-axis CCL cryogenic magnetometer surrounded by a set of Helmholtz coils and a μ-metal shield, or on a MOLSPIN spinner magnetometer. Both magnetometers are interfaced to a microcomputer system enabling analytical errors to be evaluated simultaneously with the remanence vector. Thermal and alternating field (AF) demagnetization experiments have been carried out in a magnetically shielded furnace and AF demagnetizer (ambient field <15 nT). Demagnetization results from each specimen were plotted on orthogonal vector diagrams (Zijderveld 1967) as well as on equal-area projections. A computerized line and plane fitting procedure (LINEFIND; Kent, Briden & Mardia 1983) was then used to identify linear and/or planar segments obtained during stepwise demagnetization. Only linear elements based on at least stepwise demagnetization. Only linear elements based on at least four consecutive data points were accepted as being reliable estimates for any component of magnetization removed during demagnetization.

Results of detailed pilot studies proved AF demagnetization to be generally less effective in separating the different components of magnetization than thermal demagnetization, so the bulk of the collection was subjected to stepwise thermal demagnetization.

4 PALAEOMAGNETIC RESULTS

4.1 Identification of remanence components

During the demagnetization experiments three significantly different directional components, labelled **A**, **B**, and **C** have been identified (Table 1 and Figs 3–8).

Component **A** parallels the direction of the present field of the geocentric axial dipole (PDF), or its antipodal direction, and is believed to be of Recent or Pleistocene origin.

Component **B** differs significantly from the direction of the PDF and points to the north and up or to the south and down.

Both components **A** and **B** cover the whole range of the blocking temperature spectrum up to 680°C, whenever either of them is the sole or predominant carrier of magnetization. In samples where the third component **C** could be identified, however, **A** and **B** occupy the low and intermediate blocking temperature ranges and are readily removed by heating up to 400 °C.

Component **C** generally points towards the west and down and is characterized by steep inclinations. Maximum blocking temperatures for this component do not exceed 580 °C.

4.2 A component

Component **A**, pointing to the north and down (or in a minority of cases south and up) is the most prevalent component, and has been identified in about 50 per cent of the collection. It is carried by haematite, as indicated by blocking temperatures exceeding 580 °C (Fig. 3a), maghaemite as demonstrated by the loss of more than 50 per cent of magnetization at about 350° to 400 °C in combination with low coercivities, or by both magnetic phases coexisting (Fig. 3b). Normal and reversed polarities of component **A** have been identified (Fig. 3c, d and Table 1). The proximity of the site mean directions of component **A** (Table 1 and Fig. 3d) to the local dipole field axis (*Dec./Inc.* = 0°/32.5°), however, makes an ancient origin of this component highly unlikely. The remanence carrying haematite and maghaemite are both interpreted to be the products of low-temperature oxidation of magnetite probably during Pleistocene and/or Recent deep-reaching weathering.

4.3 B component

The rockmagnetic properties of samples dominated by the presence of component **B** pointing to the north and up (Fig. 4a) or south and down (Fig. 4d), are not discernibly different from those carrying component **A**. Maximum blocking temperatures of 680 °C (Fig. 4b) and very high coercivities as demonstrated by isothermal remanent magnetization (IRM) acquisition experiments (Fig. 4c) unequivocal identify haematite as one carrier of this magnetization. As for component **A**, there is again evidence that component **B** also resides in a phase exhibiting rather low coercivity as well as intermediate maximum blocking temperatures. Intensity loss of up to 90 per cent upon thermal demagnetization to 400 °C (Fig. 4e) in combination with rather low coercivities (Fig. 4f), signify maghaemite as

Table 1. Site mean directions from the Gilif Hills, Sudan (17.83°N, 32.67°E).

	A-Component				B-Component				C-Component			
Site	N/n	Dec/ Inc	k	α_{95}	N/n	Dec/ Inc	k	α_{95}	N/n	Dec/ Inc	k	α_{95}
AL1	6/5	2.8/ +37.7	92.3	8.0	6/1	168.9/+54.0	-	-	6/1	246.6/ +57.8	-	-
JA1	8/0	-/ -	-	-	8/1	216.6/ +44.2	-	-	8/5	278.8/ +63.8	17.7	18.7
JA2	4/3	358.6/ +20.4	65.9	15.4	4/0	-/ -	-	-	4/0	-/ -	-	-
JA3	4/4	2.4/ +38.2	80.2	10.2	4/0	-/ -	-	-	4/0	-/ -	-	-
JA4	5/5	2.0/ +27.7	119.9	7.0	5/3	355.9/ -28.0	279.1	7.2	5/2	41.6/ +77.6 [1]	-	-
JA5	7/0	-/ -	-	-	7/0	-/ -	-	-	7/3	236.9/ -71.3	62.8	15.7
JAA3	7/0	-/ -	-	-	7/5	186.2/ +44.7	36.6	12.8	7/4	280.4/ +58.9	46.9	13.6
JAB3	4/0	-/ -	-	-	4/2	185.0/ +39.7	-	-	4/2	301.8/ + 78.1	-	-
NU1	4/3	353.6/ +22.7	98.0	12.5	4/0	-/ -	-	-	4/0	-/ -	-	-
NU2	8/3	0.1/ +33.7	33.5	21.6	8/2	189.0/ +38.0	-	-	8/2	358.5/ +74.6 [1]	-	-
NU3	6/4	6.3/ +28.8	41.1	14.5	6/1	4.5/ -42.2	-	-	6/2	207.0/ -75.7	-	-
NU4	7/4	2.3/ +41.7 [1]	36.5	15.4	7/0	-/ -	-	-	7/7	278.3/ +69.2	135.2	5.2
NU5	6/5	184.3/ -34.3	42.1	11.9	6/0	-/ -	-	-	6/0	-/ -	-	-
US1	11/4	180.3/ -16.3	40.8	14.6	11/2	188.4/ +33.7	-	-	11/6	174.1/ +76.5	33.3	11.8
US2	8/3	4.0/ +33.4	47.4	18.1	8/4	358.2/ -37.0 [1]	47.2	13.5	8/0	-/ -	-	-
US3	10/0	-/ -	-	-	10/1	14.7/ -37.4	-	-	10/5	298.7/ +71.8	19.4	17.8
US4	4/3	9.4/ +28.8	39.3	19.8	4/0	-/ -	-	-	4/0	-/ -	-	-
US5	5/0	-/ -	-	-	5/0	-/ -	-	-	5/0	-/ -	-	-
Overall Means:												
	12	0.9/ +29.8	91.1	4.6	10	6.5/ -40.2	48.1	7.0	11	296.2/ +79.3	18.7	10.8

Palaeo-south pole positions:

Lat. [°N]	Long. [°E]	dp	dm	Lat. [°N]	Long. [°E]	dp	dm	Lat. [°N]	Long. [°E]	dp	dm
88.0	7.8	2.8	5.1	-48.8	23.5	5.1	8.4	25.9	11.6	19.6	20.6

N/n are the number of samples measured and the number of samples used to compute the site mean direction. Dec, Inc: mean declination, inclination; k: precision parameter; α95: semi angle of the cone of confidence about the mean direction at the 95% confidence level (Fisher, 1953). dp,dm: semi axes of the oval of 95% confidence about the mean pole. Lat.,Long.: latitude,longitude of the palaeomagnetic pole.

[1] Mixed polarities within site

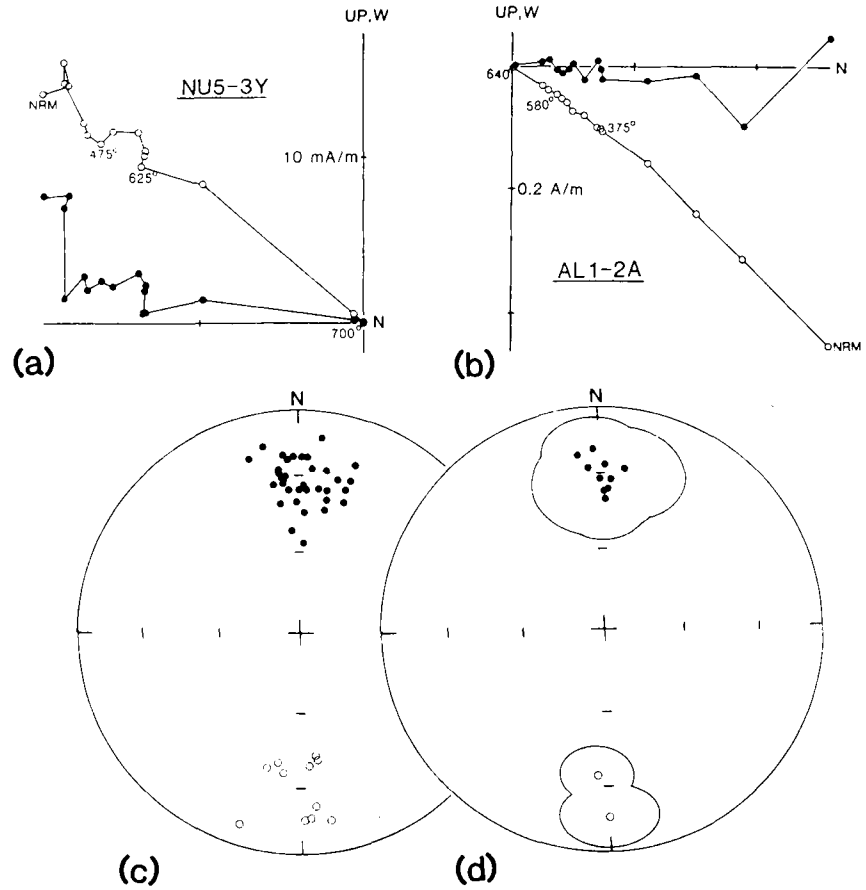


Figure 3. Component A in orthogonal demagnetization diagrams (a, b). The characteristic directions (c) for this component and the resulting site mean directions with α_{95} confidence limits (d) are also shown in equal area projection. In (a) and (b) the vertical projection of the magnetization is shown as open circle and the horizontal projection by a full circle. Open and closed circles on (c) and (d) represent projections onto the upper and lower hemispheres respectively.

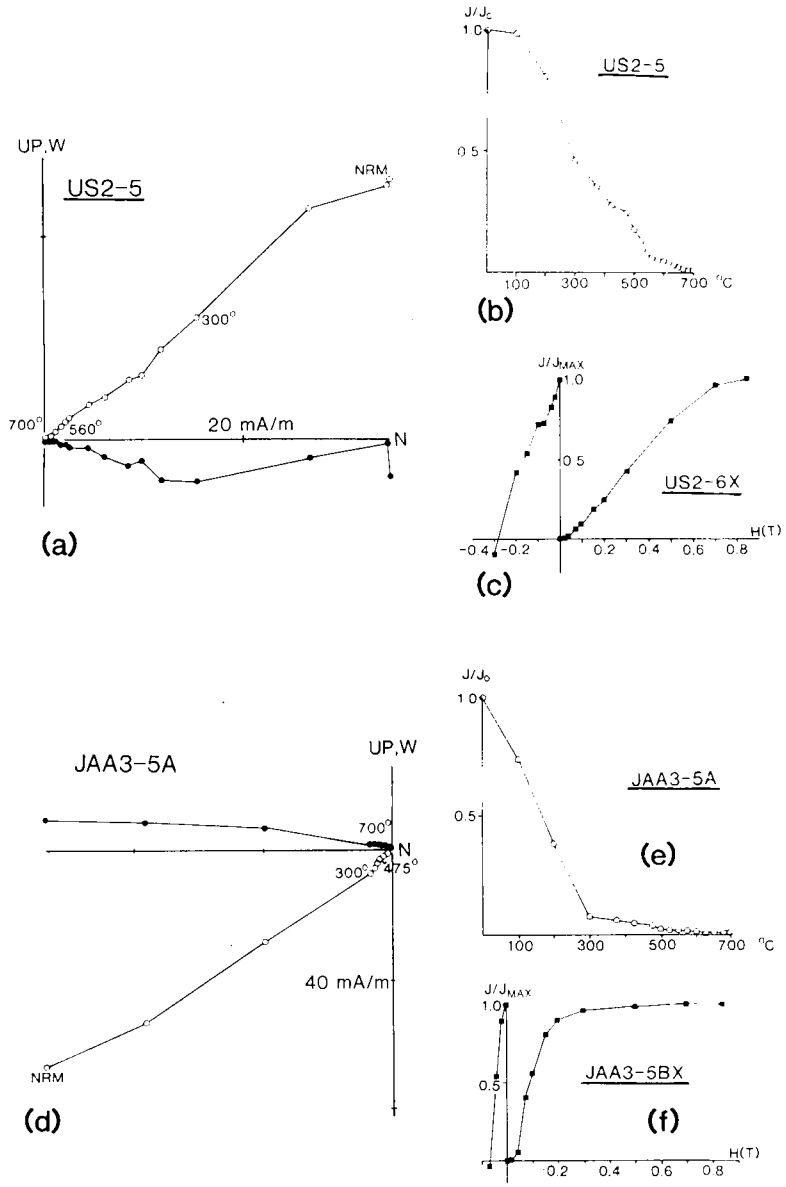


Figure 4. Typical demagnetization diagrams for samples dominated by component **B** (a, d). Conventions as in Fig. 3. Also shown are the thermal demagnetization curves (b, e) and the results of isothermal remanent magnetization acquisition experiments (c, f).

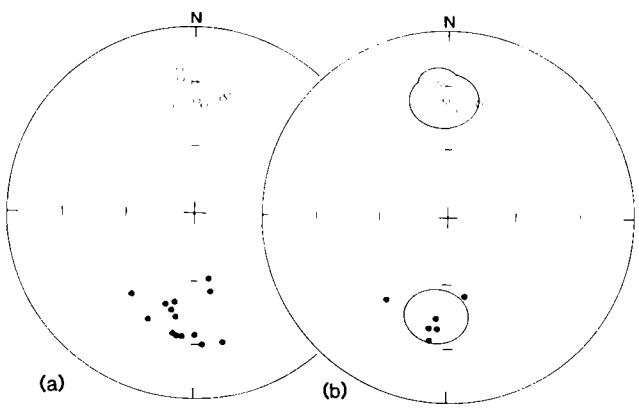


Figure 5. Equal area projections of the characteristic directions of component **B** at sample (a) and site level (b). Conventions as in Fig. 3.

the carrier of component **B** in a few samples. Component **B** was identified in a total of 22 samples from 10 sites (Table 1 and Fig. 5a, b).

4.4 C component

Typical demagnetization diagrams for samples in which component **C** was identified together with the results of rockmagnetic experiments are shown in Figs 6 and 7. Demagnetization of samples carrying component **C** was for the most part straightforward. Thermal demagnetization up to 100 °C was generally successful in removing a viscous overprint of varying magnitude. Further demagnetization of up to 375 °C successfully removed the northerly and downward pointing component **A** (Fig. 6a). Above this temperature a very steep westerly directed and downward pointing component of magnetization could be unam-

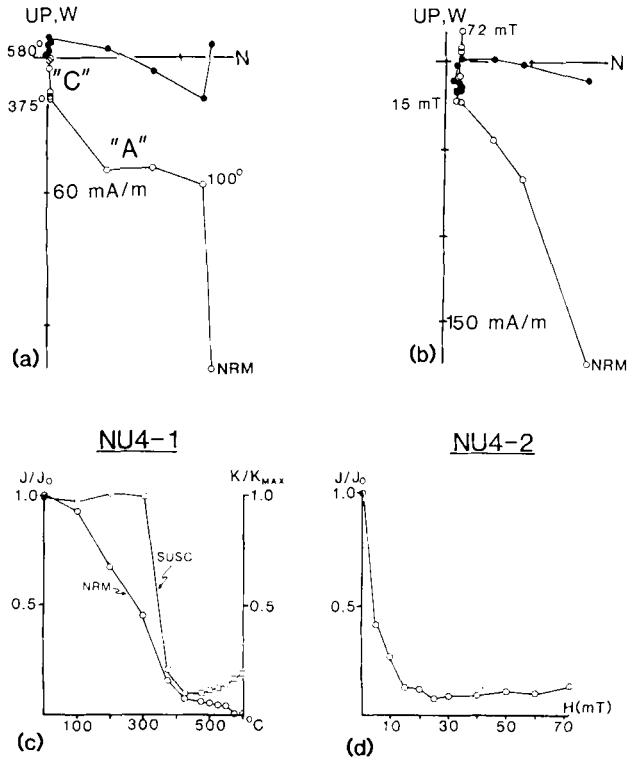


Figure 6. Thermal (a) and AF (b) demagnetization of samples carrying component **C**. The decay of intensity as a function of demagnetization temperature together with the change in bulk susceptibility is shown in (c). (d) shows the decay of intensity during AF demagnetization. Notations as in Fig. 3.

biguously identified (Fig. 6a). Maximum blocking temperatures did not exceed 580 °C (Fig. 6c). In addition, it is noteworthy that the directional change from component **A** to **C** between 350° and 375 °C (NU4–1, Fig. 6a) is generally accompanied by a marked drop in bulk susceptibility (Fig. 6c). This is believed to indicate the inversion of maghaemite to haematite during thermal demagnetization. As one would expect, AF demagnetization, though often successful in resolving component **C**, proved less efficient in resolving the complete directional spectrum whenever the presence of a high-coercivity phase was indicated (Fig. 6d). In only a small minority of the specimens did all three components **A**, **B**, and **C** exist in the same specimen (Fig. 7a). Separate blocking temperature ranges (Fig. 7b) facilitate the identification of components **A**, **B**, and **C** as linear segments (Fig. 7a). Heating to 400°–425 °C destroys about 80 per cent of the total magnetization by removing component **A**. Component **B**, pointing to the north and up is then subsequently removed and reaches maximum blocking temperatures of about 550 °C. Again the change in direction is associated with a decrease in bulk susceptibility (Fig. 7c) although this intensity drop is less pronounced than in samples dominated by maghaemite only. Above 550 °C, component **C** can be identified as a linear segment decaying towards the origin of the diagram. The removal of a direction reversed to the direction of component **C** is demonstrated in sample JA5–6 (Fig. 7d, e). The site mean directions for component **C** are shown in Fig. 8 and listed in Table 1.

Although component **C** is generally identified by linear segments, slightly curved decay trajectories (e.g. Fig. 7d) are also occasionally encountered, suggesting the possibility of partial contamination of component **C** with lower blocking temperature components. Whenever possible, additional great circle analysis was carried out in order to obtain further information on the accuracy of component **C**. 21 samples from sites where component **C** was identified yielded planar elements for the upper part of the blocking temperature spectrum, which showed high enough precision to be used in great-circle analysis. The resulting great-circle intersection (although associated with a rather high α_{95} of 26°) lies at 285°/69° and is not significantly different from the mean direction obtained for component **C**, exclusively based on linear elements (Table 1). Great-circle analysis therefore provides strong additional support for our contention that component **C** is indeed well isolated.

5 INTERPRETATION

5.1 Age of the magnetization

Based on the fact that the direction component **A** is virtually indistinguishable from the direction of the PDF at the Gilif Hills and on the rockmagnetic data a recent origin of component **A** has to be assumed. The occurrence of directions antipodal to **A** (Fig. 3a, b) extends the time interval of remagnetization back into the Pleistocene.

The age assignment for components **B** and **C** is less clear. The fact that component **B** seems to be associated with either haematite and/or maghaemite manifests the low-temperature oxidation (chemical) origin of this magnetization and points towards large-scale post-emplacement regional alteration of the Gilif Hills volcanic rocks. Alteration of the magnetic and non-magnetic minerals is also evident under the microscope, but seems to vary significantly from sample to sample, samples carrying component **C** being the least affected and displaying evidence for deuteric oxidation. The high blocking temperature and the inference that magnetite is the carrier of component **C** suggests it pre-dates component **B**. However in the absence of decisive field tests, the timing of acquisition of the NRM is intrinsically difficult to assess.

A set of selected samples has been subjected to a modified Thellier–Thellier experiment (McClelland–Brown 1982) in order to establish the nature of the three components, and in particular to distinguish between chemical remanent magnetization (CRM) of probably secondary origin and primary thermo-remanent magnetization (TRM). After heating in a controlled field of 42 μT and subsequent measuring, the samples were demagnetized at the same temperature in field free space and measured again. This cycle was repeated at progressively increasing temperatures. The NRM decay curve was then calculated for a multidirectional model using the Kent *et al.* (1983) algorithm. Vector subtraction was used to rebuild the true TRM decay curve. The TRM and NRM decay curves for samples carrying only component **A** do not show any similarities. The results for samples carrying solely component **B** were inconclusive. However, very good agreement between NRM and TRM decay curves could be achieved for samples in which component **C** was identified

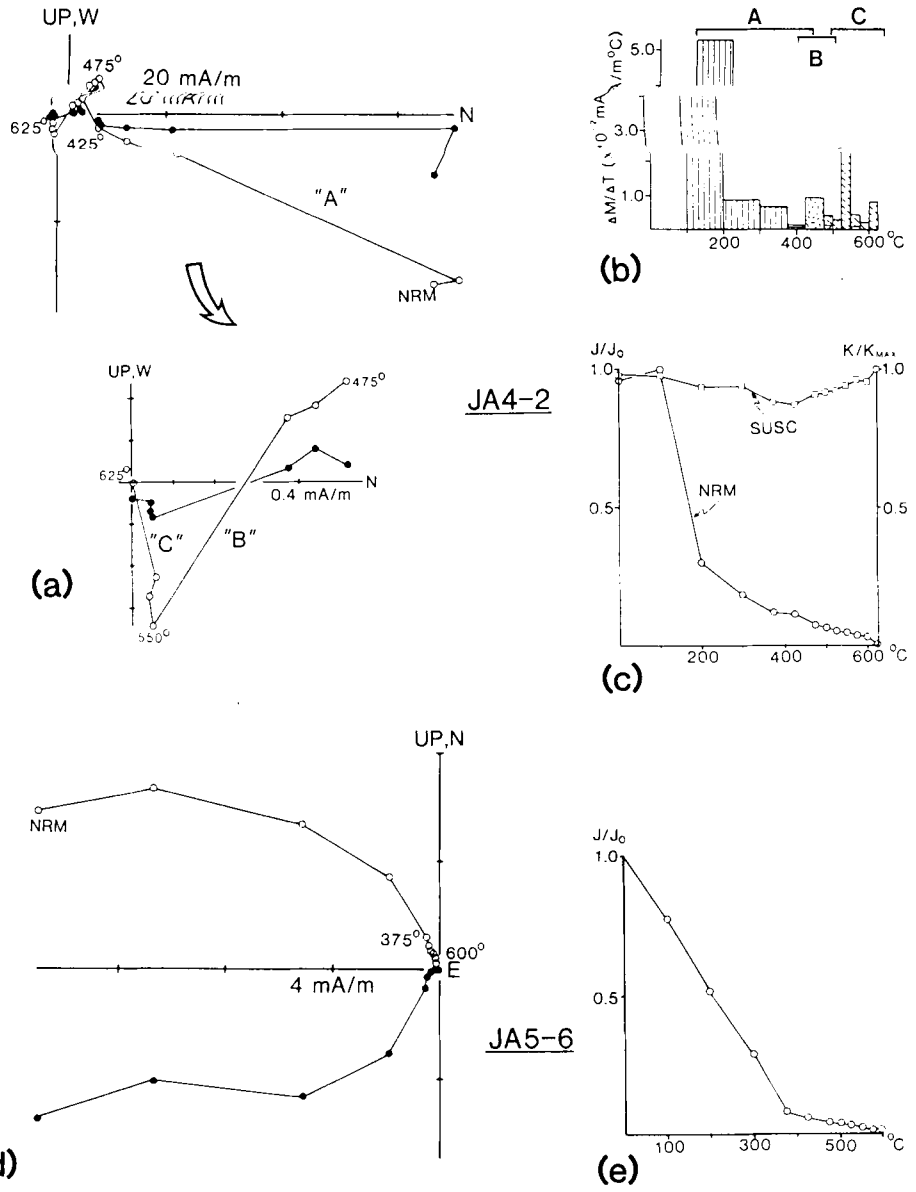


Figure 7. Orthogonal projection of results of stepwise thermal demagnetization of a sample carrying components **A**, **B** and **C** (a) and a sample carrying a reversed component **C** (d). Also shown are the stability spectrum of the individual components (b), the thermal demagnetization curves (c, e) and change in bulk susceptibility during thermal demagnetization (c). Notations as in Fig. 3.

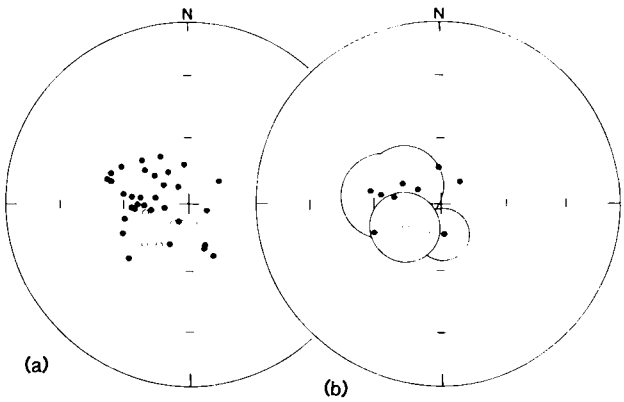


Figure 8. Equal area projections of the characteristic directions of component **C** at sample (a) and site level (b). Conventions as in Fig. 3.

as being the dominating NRM (Fig. 9). Normalizing to the intensity measured after heating at 425 °C, the temperature at which both components **A** and **B** have been removed, results in a almost perfect match of the higher temperature sections of the two curves (Fig. 9b). This suggests that the NRM removed up to 425° is very unlikely to be a TRM and, in the absence of plausible alternative mechanisms, a chemical origin of **A** and **B** can be assumed. However, the excellent match between the NRM and TRM curves above 425 °C and up to 580 °C may indicate that component **C** represents a TRM (i.e. a primary magnetization). Changes in bulk susceptibility during the experiment are ascribed to the inversion of maghaemite to haematite, which does however not affect the blocking temperature spectrum of minerals unblocking at or below the maximum unblocking temperature for magnetite. Hence rockmagnetic analysis and the directional behaviour of the NRM during stepwise

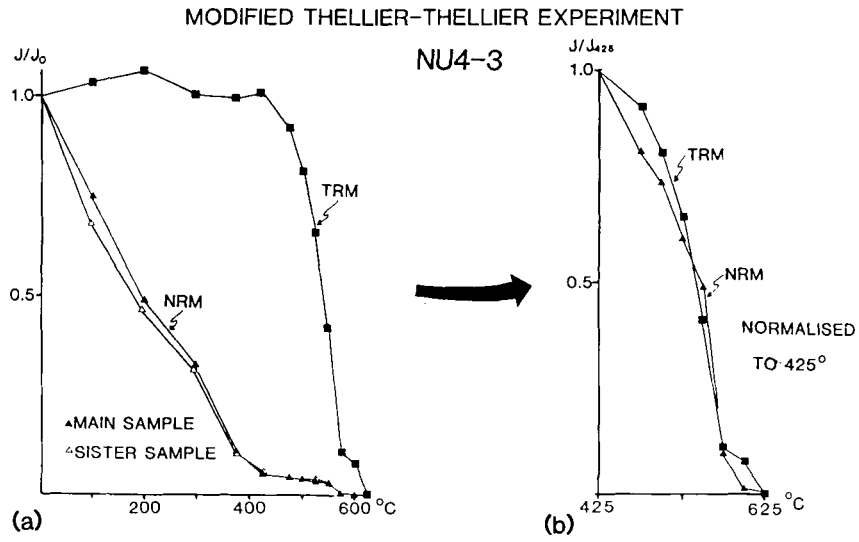


Figure 9. Results of a modified Thellier–Thellier experiment on a sample dominated by component C. Shown are the decay of the NRM (open and closed triangles) as well as the decay of the laboratory TRM (closed squares). The open triangles in (a) represent the decay of NRM of a sister sample demagnetized under field free conditions only, indicating that the NRM decay curve obtained during the Thellier–Thellier experiment has not been contaminated by TRM effects. In (b) both TRM and NRM curves have been normalized to 425 °C.

Table 2. Selected Silurian to Permo-Carboniferous palaeopole positions from Africa, Madagascar and Australia in a Gondwana Reassembly[1].

Formation	Symbol	age[2]	Lat. [°N]	Long. [°E]	k	α95 (dp,dm)	Reference
Africa							
Air intrusives, Niger	AI	S 435(?)[3]	-43	9	50	6	Hargraves et al., 1987
Gilif Hills volcanics C, Sudan	GC	Dm377±5[4]	26	12	19	11	this study
Bokkeveld Group, South Africa	BO	Dm[5](370)	10	15	-	(9,11)	Bachtadse et al., 1987
Gilif Hills volcanics B, Sudan	GB	Cu[5](280)	-49	23	56	6	this study
Dwyka varves, Sambia	DW	Cu[5](310)	-26	26	52	10	McElhinny and Opdyke, 1968
Hassi-Bechar red beds, Algeria	HC	Cu[5](280)	-27	57	-	(2,4)	Daly and Irving, 1983
Abadla Red Beds, Algeria	AB	Cu[5](280)	-29	60	-	(3,6)	Morel et al., 1981
Ain Ech Chebbi, Algeria	AC	Cu[5](280)	-23	52	123	6	Daly and Irving, 1983
Galula redbeds, Tanzania	GA	P-C[5](280)	-46	40	11	16	McElhinny et al., 1968
Songwe-Ketawaka red beds, Tanzania	SO	Pl[5](275)	-27	89	-	16	McElhinny et al., 1968
Chougrane red beds, Morocco	CH	Pl[5](275)	-32	61	-	5	Daly and Pozzi, 1976
Jebel Tarhat red beds, Morocco	TA	Pl[5](275)	-24	64	-	6	Martin et al., 1978
Hassi-Bechar Permian, Algeria	HP	Pl[5](270)	-36	59	-	(3,5)	Daly and Irving, 1983
Taztot andesite, Morocco	TZ	Pl[5](270)	-39	57	-	4	Daly and Pozzi, 1976
Madagascar							
Sakoa Group glacials	SK	Cu[5](285)	-43	60	50	8	McElhinny and Embleton, 1976
Sakamena Group	SA	Pu[5](240)	-63	72	79	4	McElhinny and Embleton, 1976
Australia							
Snowy river volcanics[7]	SV	Dl[5](400)	-54	15	-	(11,14)	Schmidt et al., 1986
Mugga Mugga porphyry[7]	MP	S 414±8[4]	-16	2	-	(6,9)	Briden, 1966; revised by Goleby, 1980
Hervey Group	HE	Du[5](365)	-5	8	-	(8,16)	Li et al., 1988
Canning Basin	CB	Du[5](370)	2	16	62	8	Hurley and Van der Voo, 1987
Comerong volcanics[7]	CV	Dm[5](380)	-34	2	42	7	Schmidt et al., 1987

α_{95}: semi angle of the cone of confidence about the palaeopole position at the 95% probability level. dp,dm: semi axis of the oval of 95% confidence about the mean pole. Lat.,Long.: latitude,longitude of the palaeomagnetic pole. k is the Fisherian precision parameter associated with the means. D: Devonian; P: Permian; C: Carboniferous; l, m, u: lower, middle, upper.

1) Palaeomagnetic poles from Australia and Madagascar have been transferred into African coordinates using the rotation parameters given by Lawver and Scotese (1987).
2) numbers in brackets are estimates of the absolute age, based on the stratigraphic information and the Harland et al. (1982) time scale.
3) Ar^{40}/Ar^{39} age
4) Rb/Sr age
5) stratigraphic age
6) probable remagnetization age
7) palaeopole positions from the (allochthonous?) eastern Australian Lachlan Fold Belt, not used to compute APW path in Fig. 11.

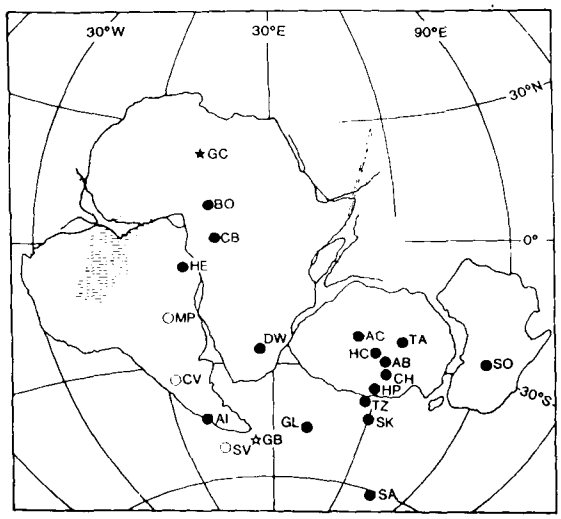


Figure 10. Resulting palaeo-south pole positions for component **B** (open star) and **C** (solid star) plotted on a reconstruction of Gondwana (Lawver & Scotese 1988). Also shown are Devonian, Carboniferous and Permian palaeopoles from Africa and Australia (full circles). Poles from the (allochthonous?) eastern Australian Lachlan Fold Belt are shown as open circles. Refer to Table 2 for symbols. Stippled areas mark the known occurrence of Devonian tillites in Africa and South America (after Caputo & Crowell 1985).

thermal and AF demagnetization experiments lead us to believe that the magnetization of component **C** was acquired during or shortly after the closure of the Rb/Sr system at 377 ± 5 Myr (Barth *et al.* 1983). Based on the rockmagnetic observations, a secondary origin of component **B** is indicated. According to the position of the resulting palaeopole for component **B** the age of remagnetization, can be bracketed between the earliest Carboniferous (pole DW in Table 2 and Fig. 10) and the cluster of palaeopoles of early Permian age, situated to the east of southern Africa (Fig. 10). The position of pole GB is virtually identical to palaeomagnetic results obtained from two syenite intrusions from the eastern Sudanese Red Sea Hills dated at 576 ± 23 Myr (K/Ar, Hornblende) and 468 Myr (K/Ar, biotite) (D. Rex, personal communication), which have also been interpreted to reflect late Palaeozoic remagnetizations (V. Bachtadse, unpublished data). In addition, the pole for component **B** is in good agreement with a Carboniferous pole position based on palaeoclimatological data (Scotese & Barrett 1990).

5.2 Consequences for the African apparent polar wander path

The evidence that component **C** may be TRM, the identification of magnetite as its carrier and the fact that its direction does not match any expected post-Devonian direction, all lend support to our interpretation of component **C** as a primary mid-Devonian magnetization. The palaeomagnetic south pole for the Devonian as reported here (GC; Fig. 10) plots between a tight cluster of Ordovician south poles to the northwest of North Africa (Bachtadse & Briden 1990) and a group of palaeo-south pole positions for the mid to latest Devonian of Africa and Australia (Fig. 10; Table 2) namely for the Bokkeveld Group (BO; Bachtadse *et al.* 1987), the Canning Basin (CB; Hurley & Van der Voo 1987) and the Hervey Group (HE; Li, Schmidt & Embleton 1988). In addition it is noteworthy that those poles plot rather close to the occurrence of glacial deposits of Late Devonian age (Fig. 10) in the Amazon (Rocha-Campos 1981a), Paranaiba (Rocha-Campos 1981b) and Solimões (Caputo & Crowell 1985) basins of South America and similar occurrences of less well-constrained age in the Tasmena (Hambrey & Kluyer 1981) and Accra (Talbot 1981) basins of western Africa (Fig. 10). The predicted palaeolatitude based on pole GC puts the margin of Northern Africa in rather high southern latitude of approx. 80° during the early to mid-Devonian. Comparing this result with recently published palaeogeographic reconstructions for Laurentia (Torsvik *et al.* 1990) yields a latitudinal separation between Northern Africa and the northern continents of up to 50°, thus indicating the existence of a major pre-Hercynian ocean basin, at least 5000 km in width, in the Devonian.

Although we do not have direct evidence for the age of **B** magnetization a Carboniferous to early Permian age is an inescapable conclusion, based on comparison of the *latitudinal* position of the resulting pole (GB) to other late Palaeozoic and younger palaeomagnetic data for Africa (Fig. 10 and Table 2). Pole GB is some 30° of arc west of the majority of the late Palaeozoic poles. Asuming that the Nubian shield was an integral part of Africa by Devonian time, a westward diversion of the Carboniferous APW path is indicated (Fig. 11). This corresponds to an anti-clockwise rotation of the whole of Africa and, by inference Gondwana. Such a rotation is also indicated by the large-scale late orogenic (late Carboniferous–early Permian) dextral strike slip in Hercynian Europe and North Africa (Arthaud & Matte 1977). The GB pole is very close to the Aïr pole (AI). Therefore the age of the Aïr pole has to be reassessed. Re(?)-magnetization cannot be ruled out and an APW path without a loop (Fig. 11) becomes viable.

This interpretation (Fig. 11), does not reconcile palaeomagnetic data for the Palaeozoic from eastern Australia with the APW path for Africa. Li *et al.* (1990)

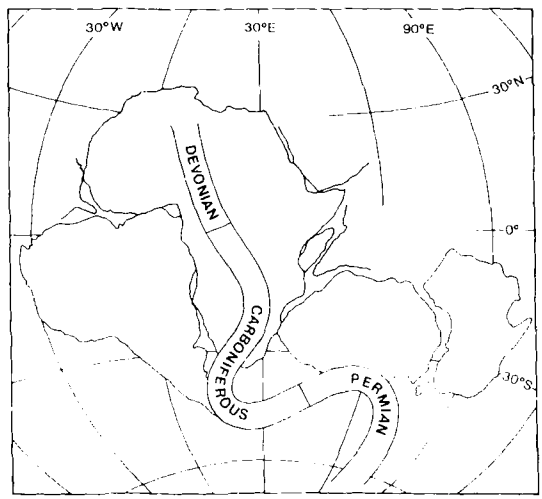


Figure 11. Apparent polar wander path for Gondwana for Devonian to Permian times based on the data listed in Table 2. Fitted by the method of Jupp & Kent (1987).

have recently interpreted the Australian data with an even larger (180°) and more sharply backtracking APW loop than the African version in Fig. 1(b), even though almost no palaeomagnetic poles are dated to within one geological epoch.

Clearly such enormous drift of Gondwana, and postulates of closure and opening of large oceans that are not prominent on geological evidence, cannot be sustained on such uncertain grounds. It becomes of paramount importance to determine the ages of remanences in the several key formations in Africa and Australia with great accuracy. Meanwhile the conservative solution presented in this paper (Fig. 11) is to be preferred.

ACKNOWLEDGMENTS

This study was supported by a British Petroleum Extramural Research Award to JCB. Fieldwork was conducted by one of us (VB) with A. J. Mortimer who, with Bushra Ibrahim, was an invaluable help in the field. The generous support of Robertson International in Sudan, particularly Dr D. S. Wood, C. Cornwell and P. Boswell is gratefully acknowledged. We are indebted to the Geological Research Authority of Sudan for granting us the permits to work in the Bayuda desert and for invaluable logistical support. Our thanks are extended to D. P. Lowe who carried out most of the rockmagnetic and palaeomagnetic measurements with never ending enthusiasm. Finally we wish to thank Drs A. Smethurst, T. Torsvik and A. Trench, and referees R. B. Hargraves and R. Van der Voo for helpful and constructive criticism.

REFERENCES

Aïfa, T., Feinberg, H. & Pozzi, J. P., 1990. Devonian/Carboniferous paleopoles for Africa. Consequences for Hercynian geodynamics, *Tectonophysics,* **179,** 288–304.

Arthaud, F. & Matte, P., 1977. Late Paleozoic strike slip faulting in southern Europe and northern Africa: Result of a right-lateral shear zone between the Appalachians and the Urals, *Geol. Soc. Am. Bull.*, **88,** 1305–1320.

Bachtadse, V. & Briden, J. C., 1990. Palaeomagnetic constraints on the position of Gondwana during Ordovician to Devonian Times, in *Palaeozoic Palaeogeography and Biogeography,* pp. 43–48, eds McKerrow, W. S. & Scotese, C. R., Geological Society, London.

Bachtadse, V., Van der Voo, R. & Haelbich, I. W., 1987. Paleomagnetism of the Western Cape Fold Belt, South Africa, and its bearing on the Paleozoic apparent polar wander path for Gondwana, *Earth planet. Sci. Lett.*, **84,** 487–499.

Barth, H., Besang, C., Lenz, H. & Meinhold, K.-D., 1983. Results of petrological investigations and Rb/Sr age determinations on the non-orogenic igneous ring-complexes in the Bayuda desert, Sudan, *Geol. Jb.*, **B51,** 3–34.

Bowden, P., van Breemen, O., Hutchinson, J. & Turner, D. C., 1976. Palaeozoic and Mesozoic age trends for some ring complexes in Niger and Nigeria, *Nature,* **259,** 297–299.

Briden, J. C., 1966. Estimates of direction and intensity of the palaeomagnetic field from the Mugga Mugga Porphyry, Australia, *Geophys. J. R. astr. Soc.*, **11,** 267–278.

Brock, A., 1981. Paleomagnetism of Africa and Madagascar, in *Paleoreconstruction of the Continents,* pp. 65–76, eds McElhinny, M. W. & Valencio, D. A., AGU, Washington, DC.

Caputo, M. V. & Crowell, J. C., 1985. Migration of glacial centres across Gondwana during the Paleozoic Era, *Geol. Soc. Am. Bull.*, **96,** 1020–1036.

Daly, L. & Irving, E., 1983. Paleomagnetism des roches carboniferes du Sahara central; analyse des aimantations juxtaposées; configuration de la Pangée, *Ann. Geophys.*, **1,** 207–216.

Daly, L. & Pozzi, J.-P., 1976. Résultats paléomagnetiques du Permien inférieur et du Trias marocain: comparaison avec les données Africaines et Sud Americaines, *Earth planet. Sci. Lett.*, **29,** 71–80.

Fisher, R. A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, A, **217,** 295–305.

Goleby, B. R., 1980. Early Paleozoic paleomagnetism in South East Australia, *J. Geomag. Geoelectr.*, **32,** SIII 11–SIII 21.

Hailwood, E. A., 1974. Palaeomagnetism of the Msissi Norite (Morocco) and the Palaeozoic reconstruction of Gondwana, *Earth planet. Sci. Lett.*, **23,** 376–386.

Hambrey, M. J. & Kluyer, H. M., 1981. Evidence of Devonian or Early Carboniferous glaciation in the Agades region of Niger, in *Earth's Pre-Pleistocene Glacial Record,* pp. 188–190, eds Hambrey, M. J. & Harland, W. B., Cambridge University Press, Cambridge, UK.

Hargraves, R. B., Dawson, E. M. & Van Houten, F. B., 1987. Palaeomagnetism and age of mid Palaeozoic ring complexes in Niger, West Africa, and tectonic implications, *Geophys. J. R. astr. Soc.*, **90,** 705–729.

Harland, W. B., Cox, A. V., Llewellyn, P. G., Pickton, C. A. G., Smith, A. G. & Walters, R., 1982. *A Geologic Time Scale,* Cambridge University Press, Cambridge, UK.

Hurley, N. F. & Van der Voo, R., 1987. Paleomagnetism of Upper Devonian reefal limestones, Canning Basin, Western Australia, *Geol. Soc. Am. Bull.*, **98,** 138–146.

Jupp, P. E. & Kent, J. T., 1987. Fitting smooth paths to spherical data, *Appl. Stat.*, **36,** 34–36.

Kent, D. V., Dia, O. & Sougy, J. M. A., 1984. Paleomagnetism of Lower-Middle Devonian and Upper Proterozoic–Cambrian(?) rocks from Mejeria (Mauritania, West Africa), in *Plate Reconstructions from Paleozoic Paleomagnetism,* Geodynamic Series, vol. 12. pp. 99–115, eds Van der Voo, R. & Scotese, C. R., AGU, Washington, DC.

Kent, J. T., Briden, J. C. & Mardia, K. V., 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence, *Geophys. J. R. astr. Soc.*, **75,** 593–621.

Kinnaird, J. & Bowden, P., 1987. African anorogenic alkaline magmatism and mineralization—a discussion with reference to the Niger–Nigerian Province, *Geol. J.*, **22,** 297–340.

Lawver, L. A. & Scotese, C. R., 1987. A revised reconstruction of Gondwana, in *Gondwana Six: Structure, Tectonics and Geophysics,* pp. 17–23, ed. McKenzie, G. D., AGU, Washington, DC.

Li, Z. X., Schmidt, P. W. & Embleton, B. J. J., 1988. Paleomagnetism of the Hervey Group, central New South Wales, and its tectonic implications, *Tectonics,* **7,** 351–367.

Li, Z. X., Powell, C. McA., Thrupp, G. A. & Schmidt, P. W., 1990. Australian Palaeozoic palaeomagnetism and tectonics—II. A revised apparent polar wander path and palaeogeography, *J. Structural Geol.*, **12,** 567–575.

Livermore, R. A., Smith, A. G. & Briden, J. C., 1985. Palaeomagnetic constraints on the distribution of continents in the late Silurian and early Devonian, *Phil. Trans. R. Soc. Lond.*, B, **309,** 29–56.

Martin, D. L., Nairn, A. E. M., Noltimier, H. C., Petty, M. H. & Schmitt, T. J., 1978. Paleozoic and Mesozoic Paleomagnetic results from Morocco, *Tectonophysics,* **44,** 81–114.

McClelland-Brown, E. 1982. Discrimination of TRM and CRM by blocking-temperature spectrum analysis, *Phys. Earth planet. Inter.*, **30,** 405–414.

McElhinny, M. W. & Opdyke, N. D., 1968. The paleomagnetism of some Carboniferous glacial varves from Central Africa, *J. geophys. Res.*, **73,** 689–696.

McElhinny, M. W. & Embleton, B. J. J., 1976. The palaeomagnetism of Madagascar: remanence and magnetic properties of Late Palaeozoic sediments, *Earth planet. Sci. Lett.*, **31,** 101–112.

McElhinny, M. W., Briden, J. C., Jones, D. L. & Brock, A., 1968. Geological and geophysical implications of paleomagnetic results from Africa, *Rev. Geophys.*, **6,** 201–238.

Morel, P. & Irving, E., 1978. Tentative paleocontinental maps for the early Phanerozoic and Proterozoic, *J. Geol.*, **86,** 535–561.

Morel, P., Irving, E., Daly, L. & Moussine-Pouchkine, A., 1981. Paleomagnetic results from Permian rocks of the northern Saharan craton and motions of the Moroccan Meseta and Pangea, *Earth planet. Sci. Lett.*, **55,** 65–74.

Rocha-Campos, A. C., 1981a. Late Devonian Curua Formation, Amazon Basin, Brazil, in *Earth's Pre-Pleistocene Glacial Record*, pp. 888–891, eds Hambrey, M. J. & Harland, W. B., Cambridge University Press, Cambridge, UK.

Rocha-Campos, A. C., 1981b. Middle-Late Devonian Cabecas Formation, Paranaiba Basin, Brazil, in *Earth's Pre-Pleistocene Glacial Record*, pp. 892–895, eds Hambrey, M. J. & Harland, W. B., Cambridge University Press, Cambridge, UK.

Salmon, E., Edel, J. B., Pique, A. & Westphal, M., 1988. Carboniferous palaeomagnetic investigations in Morocco: Permian remagnetizations and possible large Carboniferous rotations occurring in Mesetian sedimentary and Jebilets intrusive rocks, *Geophys. J. R. astr. Soc.*, **93,** 115–125.

Salmon, E., Montigny, R., Edel, J. B., Pique, A., Thuizat, R. & Westphal, M., 1986. The Msissi Norite revisited: K/Ar dating, petrography and paleomagnetism, *Geophys. Res. Lett.*, **13,** 741–743.

Schmidt, P. W., Embleton, B. J. J. & Palmer, H. C., 1987, Pre- and postfolding magnetizations from the early Devonian Snowy River Volcanics and Buchan Caves Limestone, Victoria, *Geophys. J. R. astr. Soc.*, **91,** 155–177.

Schmidt, P. W., Embleton, B. J. J., Cudahy, T. J. & Powell, C. McA., 1986. Prefolding and premagakinking magnetizations from the Devonian Comerong volcanics, New South Wales, Australia, and their bearing on the Gondwana pole path, *Tectonics*, **5,** 135–150.

Scotese, C. R. & Barrett, S. F., 1990. Gondwana's movement over the South Pole during the Paleozoic: Evidence from lithologic indicators of climate, in *Palaeozoic Palaeogeography and Biogeography*, pp. 75–85, eds McKerrow, W. S. & Scotese, C. R., Geological Society, London.

Soffel, H. C., Saradeth, S., Briden, J. C., Bachtadse, V. & Rolf, C., 1990. The Sabaloka Ring Complex revisited: Palaeomagnetism and rock magnetism, *Geophys. J. Int.*, **102,** 411–420.

Talbot, M. R., 1981. Early Palaeozoic(?) diamictites of southwest Ghana, in *Earth's Pre-Pleistocene Glacial Record*, pp. 108–112, eds Hambrey, M. J. & Harland, W. B., Cambridge University Press, Cambridge, UK.

Torsvik, T. H., Smethurst, M. A., Briden, J. C. & Sturt, B. A., 1990. A review of Palaeozoic palaeomagnetic data from Europe and their palaeogeographic implications, in *Palaeozoic Palaeogeography and Biogeography*, pp. 25–41, eds McKerrow, W. S. & Scotese, C. R., Geological Society, London.

Vail, J. R., 1976. Location and geochronology of ring-complexes and related rocks in north-east Africa, *Geol. Jb.*, **B20,** 97–114.

Vail, J. R., 1989. Ring complexes and related rocks in Africa, *J. Afr. Earth Sci.*, **8,** 19–40.

Van der Voo, R., 1988. Paleozoic paleogeography of North America, Gondwana, and intervening displaced terranes: Comparisons of paleomagnetism with paleoclimatology and biogeographical patterns, *Geol. Soc. Am. Bull.*, **100,** 311–324.

Veevers, J. J. & Powell, C. McA., 1987. Late Paleozoic glacial episodes in Gondwana reflected in transgressive/regressive depositional sequences in Euramerica, *Geol. Soc. Am. Bull.*, **89,** 475–487.

Zijderveld, J. D. A., 1967. AC demagnetization of rocks: analysis of results, in *Methods in Palaeomagnetism*, pp. 254–287, eds Collinson, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, New York.